DataMites Global Training Institute

Data Analyst Internship

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Objectives: PRSQ-02-Medical Data History

- ✓ Database Architecture & Connectivity: Establish and configure secure remote connections to RDBMS using MySQL
- ✓ Advanced Exploratory Data Analysis (EDA):Perform complex multi-table joins and data aggregations
- ✓ Clinical Quality Assurance & Risk Mitigation: Develop automated audit queries to detect data entry anomalies and ensure patient safety by verifying the alignment between specific diagnoses and attending specialists.
- ✓ Operational Intelligence & Strategic Planning: Identify cyclical "Peak Days" and the "Payday Effect" within the UAE healthcare context to provide data-backed evidence for staff

Queries:1

```
21 •  select first_name,last_name, gender from patients where gender='M';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	gender
Donald	Waterfield	M
Mickey	Baasha	M
Jiji	Sharma	M
Blair	Diaz	M
Charles	Wolfe	M
Thomas	ONeill	M
Sonny	Beckett	M
Cedric	Coltrane	M
Hank	Spencer	M
Rick	Bennett	M
Woody	Bashir	M
Tom	Halliwell	M
John	West	M
Jon	Doggett	M
Nino	Andrews	M
John	Farley	M
Jim	Thomas	M

Result Grid
Form Editor
Field Types
Query Stats

Cont..

This helps me to:

- ✓To know patients with gender
- ✓To gather information all male patients

Queries:2

```
5 •   select first_name, last_name
6     from patients
7     where allergies is null or
8     allergies='none';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name
Donald	Waterfield
Blair	Diaz
Thomas	ONeill
Sonny	Beckett
Cedric	Coltrane
Hank	Spencer
Sara	di Marco
Amy	Leela
Rachel	Winterbourne
John	West
Jon	Doggett
Angel	Edwards
Brodie	Beck
Nea	Kobayakawa
Laura	Halliwell

Queries: 3

```
8 •  select first_name from patients where first_name like 'C%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name
Charles
Cedric
Charles
Cross
Calleigh
Catherine
Caroline
Casanova
Chen
Charmian
Cary
Christine
Carol
Corrado
Cordelia
Claire
Carol

Cont..

This helps me to:

- ✓**To gather specific information of patients**

Queries: 4

```
9 •   select first_name, last_name,weight from patients where weight between 100 and 120;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	weight
Jiji	Sharma	106
Blair	Diaz	104
Thomas	ONeill	117
Sonny	Beckett	105
Tom	Halliwell	114
Jon	Doggett	116
Angel	Edwards	106
John	Farley	104
Temple	Russert	102
Don	Edwards	118
Gary	Ramotswe	101
Phil	Chester	112
Woody	Riviera	109
Rocky	Donahue	110
Shiro	Whale	117
Didi	Dean	110
Catherine	Minoru	100

Cont..

This helps me:

- To gather patients name
- **To get the patients date in specific weight range**

Queries: 5

```
11 •   select first_name, last_name, ifnull(allergies, 'NKA') as allergies from patients;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	allergies
Donald	Waterfield	NKA
Mickey	Baasha	Sulfa
Jiji	Sharma	Penicillin
Blair	Diaz	NKA
Charles	Wolfe	Penicillin
Sue	Falcon	Penicillin
Thomas	ONeill	NKA
Sonny	Beckett	NKA
Sister	Spitzer	Penicillin
Cedric	Coltrane	NKA
Hank	Spencer	NKA
Sara	di Marco	NKA
Daphne	Seabright	Codeine
Rick	Bennett	Penicillin
Amy	Leela	NKA

Result 14 × Read Only

This helps me:

- To gather patients name(first_name and last_name)
- To get patient allergies with corresponding name

Queries:6

```
12 •  select concat(first_name, ' ', last_name) as full_name from patients;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

full_name
Donald Waterfield
Mickey Baasha
Jiji Sharma
Blair Diaz
Charles Wolfe
Sue Falcon
Thomas ONeill
Sonny Beckett
Sister Spitzer
Cedric Coltrane
Hank Spencer
Sara di Marco
Daphne Seabright
Rick Bennett
Amy Leela

Result 15 × Read Only

Queries: 7

```sql
SELECT
    p.first_name,
    p.last_name,
    pn.province_name
FROM
    patients AS p
JOIN
    province_names AS pn ON p.province_id = pn.province_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	province_name
George	Magliozzi	Ontario
Emil	Owens	Ontario
Michael	Klump	Ontario
Elizabeth	Martinelli	Ontario
Ryan	Umbridge	Ontario
Kay	Rawling	Ontario
Juan	Davies	Ontario
Jaenelle	Redfield	Ontario
Ariel	Scribbs	Ontario
Byron	Briscoe	British Columbia
Robin	Calvin	Ontario
Peter	Hajime	Ontario
Maria	Aran	Ontario
Jenny	Merrova	Ontario
Lydia	Millhone	Ontario
Troy	Stone	Ontario
Annie	Carver	Ontario

Result Grid | Form Editor | Field Types | Query Stats

Result 17 | Read Only

Queries: 8

```
21 •   select count(*) as total_patients_2010 from patients where year(birth_date)=2010;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	total_patients_2010
▶	55

Queries: 9

```
22 •  select first_name, last_name, height from patients where height=(select max(height) from patients);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	first_name	last_name	height
▶	Sam	Haruko	226

Queries: 10

```
23 •  select *from patients where patient_id in(1, 45, 534, 879, 1000);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight
1	Donald	Waterfield	M	1963-02-12	Barrie	ON	NULL	156	65
45	Cross	Gordon	M	2009-03-20	Ancaster	ON	NULL	125	53
534	Don	Zatara	M	2008-01-11	Timmins	ON	NULL	136	67
879	Orla	Shawn	F	1967-09-24	Sarnia	ON	Penicillin	149	65
1000	Rick	Williams	M	1975-04-13	Hamilton	ON	Penicillin	176	127
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Queries: 11

```
24 •  select count(*) as total_number from admissions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	total_number
▶	5067

Queries: 12

```
25 •  select *from admissions where admission_date = discharge_date;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	admission_date	discharge_date	diagnosis	attending_doctor_id
1	2018-09-20	2018-09-20	Ineffective Breathin Pattern R/T Fluid Accumulatio	24
9	2018-12-31	2018-12-31	Ruptured Appendicitis	19
10	2019-02-27	2019-02-27	Lower Quadrant Pain	27
17	2019-03-04	2019-03-04	Diabetes Mellitus	9
28	2019-03-30	2019-03-30	Cancer Of The Stomach	26
31	2018-09-26	2018-09-26	Cardiovascular Disease	19
53	2018-10-24	2018-10-24	Urinary Tract Infection	8
54	2019-04-07	2019-04-07	Hypertension	21
70	2018-07-17	2018-07-17	Migraine	20
78	2018-06-17	2018-06-17	Hypertension	17
91	2018-08-30	2018-08-30	Congestive Heart Failure	3
92	2019-01-03	2019-01-03	Osteo Arthtitis Knee	5
93	2019-05-20	2019-05-20	Left Creebral Vascular Accident	12
110	2019-02-18	2019-02-18	Biliary Colic	14
111	2018-08-25	2018-08-25	Subdural Hematoma	27
113	2018-10-21	2018-1	d & 3rd Degree Burns To Right Arm / Both Legs	13
114	2019-02-11	2019-0	MVA HT	26

2018-10-21

admissions 29

Cont..

Risk factor

1. **The "Safety" Risk: Allergies (Query #5)**
2. **The "Physical" Risk: Weight & Height (Queries #4 & #9)**
3. **The "Demographic" Risk: Birth Year (Query #8)**
4. **The "Operational" Risk: Same-Day Admissions (Query #12)**

Queries: 13

```
27 •  select count(*) as total_admisions from admissions where patient_id=579;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	total_admisions
▶	2

Risk Factor

In medical data history (PRSQ-02), a high count for a single patient_id is a major Clinical Risk Factor:

1. The "Readmission" Risk (Clinical Quality):

   a. The Risk(Planned vs. Unplanned)

   b. Insight

2. The "Rising Risk" Category (Financial)

Cont..

a. Admission: Low Risk (Acute event, like a broken bone).

b. Admissions: Rising Risk. These patients are not "chronic" yet, but they are starting to use more hospital resources.

c. 10+ Admissions: High Risk (Chronic/High-Utilizer).

Insight: Identifying "2-visit" patients allows the hospital to intervene before they become high-cost chronic patients.

Cont...

3. The "Diagnostic" Risk (Medical History)

Sometimes two admissions happen because the first visit was for a symptom and the second was for a procedure.

Queries: 14

```
28 •    select distinct city from patients where province_id ='NS';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city
Port Hawkesbury
Halifax
Inverness

patients 34

Cont..

mapping unique cities serves as an Environmental Risk Assessment:

- **Resource Allocation: If you see 60 cities with province_id ‘NS’ but most patients, come from just 3, those 3 cities are "High-Risk zones" for hospital capacity.**
- **Accessibility Risk: Patients in remote cities (the "unique" outliers) are at a higher risk of missing follow-up appointments due to travel distance.**
- **Outbreak Tracking: If a specific city suddenly shows many admissions, it acts as a "Hotspot" risk factor for infectious diseases.**

Queries: 15

```
29 •  select first_name, last_name, birth_date, (weight/ (height/100) * (height/100)) as calculated_BMI
30    from patients where height>160 and weight>70;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	birth_date	calculated_BMI
Mickey	Baasha	1981-05-28	76.00000000
Jiji	Sharma	1957-09-05	106.00000000
Blair	Diaz	1967-01-07	104.00000000
Thomas	ONeill	1993-01-31	117.00000000
Sonny	Beckett	1952-12-11	105.00000000
Sister	Spitzer	1966-10-15	95.00000000
Rick	Bennett	1977-01-27	95.00000000
Amy	Leela	1977-06-25	72.00000000
Tom	Halliwell	1987-08-01	114.00000000
Rachel	Winterbourne	1966-04-26	95.00000000
Jon	Doggett	1951-12-25	116.00000000
Angel	Edwards	1975-08-22	106.00000000
Nino	Andrews	2001-04-21	85.00000000
John	Farley	1956-09-12	104.00000000
Lucille	Maureen	1953-08-22	87.00000000
Sam	Threep	1982-06-16	141.00000000
Roland	Edge	1967-09-25	134.00000000

Result 37

Cont..

In a Medical Data History analysis, combining height and weight allows me to calculate the Body Mass Index (BMI), which is one of the most significant clinical risk factors.

Queries: 16

```
32 •   select distinct year(birth_date) as birth_year from patients order by birth_year asc;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

birth_year
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016

esult 40

Cont...

In Medical Data History analysis, identifying unique birth years is the first step in Population Health Risk Assessment:

- **Aging Population Risk**
- **Vaccination Risk**

Queries: 17

```
33 •  select first_name from patients group by first_name having count(first_name)=1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name
Abby
Adelaide
Adelia
Akira
Albert
Aldo
Alec
Alicia
Allan
Alpa
Amane
Amitabh
Angel
Angelina
Angus
Anne
Anton

patients 42

Cont..

this specific query to find truly unique individuals in my database. While DISTINCT simply hides duplicates from your view, GROUP BY with HAVING COUNT = 1 actually filters out anyone who shares a name with another person.

1. **Identifying the "Lowest Risk" for Identity Errors :-In a hospital, the biggest danger is giving the wrong medicine to the wrong "John Smith."**
2. **Patient Privacy :- The Logic: If there is only one "Mekibibi" in the whole hospital, it is very easy to guess who that patient is.**

Queries:18

```
37 •  select patient_id, first_name from patients where first_name like 's%s' and  length(first_name)=6;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	first_name
496	Spiros
629	Spiros
1789	Seamus
2258	Spiros
NULL	NULL

The use of query 18

- Pattern Matching for Identifier
- Data Integrity Check

Queries: 19

```
38 •  select p.patient_id, p.first_name, p.last_name, diagnosis from patients as p
39    join admissions as a on p.patient_id=a.patient_id where diagnosis='Dementia';
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_id	first_name	last_name	diagnosis
160	Miranda	Delacour	Dementia
178	David	Bustamonte	Dementia
207	Matt	Celine	Dementia
613	Jaki	Granger	Dementia
836	Montana	Vimes	Dementia
924	Simon	Spellman	Dementia
1201	Irene	Murphy	Dementia
1264	Jillian	Valentine	Dementia
1402	Kathryn	Hallow	Dementia
1491	Doris	McGrew	Dementia
1585	Alex	Cantropus	Dementia
1749	Alejandro	Mellie	Dementia
1798	Sister	Trenton	Dementia
1844	Jadu	Wilson	Dementia
1938	Anthony	Anneke	Dementia
2292	Jessica	Talbot	Dementia
2532	Lynn	Willing	Dementia

sult 62

Key Finding:

- Management Risk: Patients with Dementia are at a High Tier(priority ranking/very high probability of something going wrong) for "Adverse Drug Events.”
  - The Problem: Cognitive impairment leads to In-hospital Polypharmacy Errors (confusion over multiple medications) and potential Double-Dosing or Omission of critical treatments.
  - The Mitigation: By querying this cohort, we trigger an immediate Pharmacist Review and "Observed Administration" protocol to ensure 100% medication compliance.

Queries: 20

```
45 •  select first_name from patients order by length(first_name) asc, first_name asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name
Al
Al
Al
Al
Al
Al
Al
Bo
Bo
Bo
Bo
Bo
Bo
Cy
Cy
Ed
Ed

patients 66

Name Analysis:

- Short Names (<3 chars): These are High-Risk Data Entry Gaps.
  - A name like “Bo" or “Cy" suggests the registrar didn't finish typing.
- Long Names (30+ chars): These are Administrative Compatibility Risks.
  - The Danger: Old hospital software often "cuts off" (truncates) long names. If the name on the medical file doesn't perfectly match the Insurance ID, the hospital won't get paid (Claim Denial).
- Alphabetical Sorting: This is a Data Cleaning Tool.
  - The Benefit: By grouping similar names together (e.g., "Mekibibi" and "Mekibibii"), you can quickly spot Duplicate Records caused by typos.

Queries: 21

```
46 •  select sum(gender='M' ) as male_count, sum(gender='F') as female_count from patients
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	male_count	female_count
▶	2468	2062

Result 67

Clinical & Operational Risk Analysis

- Resource Allocation Risk: Identifies if the hospital meets UAE Cultural Standards for separate gender-specific waiting areas and ward capacities.
- Clinical Gap Warning: A heavy gender imbalance (e.g., 80% Male) signals a High Risk of underserving maternal or gynecological health needs.
- Epidemiological Accuracy: Ensures the study of chronic conditions (e.g., Heart Disease) isn't biased, allowing for gender-specific screening priorities.

Queries: 22

```
47 •  select patient_id, diagnosis from admissions group by patient_id, diagnosis having count(*) >1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_id	diagnosis
137	Pregnancy
320	Pneumonia
1577	Congestive Heart Failure
2004	Left Shoulder Rotator Cuff Repair
2859	Severed Spine At C3
3012	Appendicitis
3367	Pyelonephritis
3468	Congestive Heart Failure
4083	Congestive Heart Failure
4121	Congestive Heart Failure
4363	Congestive Heart Failure

Repeat Diagnosis & Readmission Risk

- Treatment Failure Risk: A "Red Flag" for clinical quality. Multiple admissions for the same condition suggest the initial treatment or discharge plan was ineffective.
- Chronic Progression Risk: Signals that conditions like Dementia or Diabetes are entering an "Unstable" phase, requiring High-Tier Intervention and specialist oversight.
- Financial Penalty Risk: In the UAE, repeated admissions for the same diagnosis often trigger Insurance Claim Denials (Readmission Penalties), causing direct revenue loss for the hospital.

Queries: 23

```
48 •  select city, count(*) as total_patients from patients group by city order by total_patients desc, city asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city	total_patients
Hamilton	1938
Toronto	317
Burlington	276
Brantford	147
Ancaster	117
Stoney Creek	107
Cambridge	79
Dundas	79
Milton	65
Paris	58
Grimsby	55
Timmins	53
Ottawa	51
Halifax	50
Oakville	44
Sarnia	43
Fruitland	42

Result 69

Output

Geospatial Patient Distribution

- **Resource Scarcity Risk:** High-volume cities require more "Satellite Clinics" and specialized staff to prevent hospital overcrowding and long wait times.
- **Community Health Risk**: Sudden spikes in specific cities identify "Hotspots," allowing for targeted environmental or public health investigations.
- **Logistics & Access Risk(The "Distance" Problem)**: Identifies "Care Deserts" where patients face high transportation barriers, increasing the risk of missed chronic disease follow-ups.

Queries: 24

```
49 •  select first_name, last_name, 'patients' as role from patients union select first_name, last_name, 'doctors' as role from doctors;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	role
Morgan	McCrary	patients
Sam	OHara	patients
Charlie	Neutron	patients
Fleur	Morales	patients
Faye	Soong	patients
Christine	Turtle	patients
Armando	Matthews	patients
Karen	Costello	patients
Deanna	Barnes	patients
Molly	Hanover	patients
Constance	Yang	patients
Claude	Walls	doctors
Joshua	Green	doctors
Miriam	Tregre	doctors
James	Russo	doctors
Scott	Hill	doctors
Tasha	Phillips	doctors

Result 70

Unified Human Resource & Safety Audit

- Emergency Response: Provides a "Master List" for evacuation protocols, distinguishing between staff (Doctors) and vulnerable individuals (Patients).
- Security & Access Risk: Supports "Role-Based Access Control" (RBAC) to ensure patients do not enter high-risk or restricted medical zones.
- Infection Path Analysis: Critical for contact tracing during outbreaks to identify "Super-Spreaders" (Staff moving between wards) vs. "Isolated Risks" (Patients).

Queries: 25

```
50 •    select allergies, count(*) as total_patients from patients
51      where allergies is not null
52      group by allergies
53      order by total_patients desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

allergies	total_patients
Penicillin	1082
Codeine	305
Sulfa	157
ASA	99
Sulfa Drugs	71
Peanuts	52
Iodine	48
Tylenol	42
Bee Stings	40
Valporic Acid	38
Tetracycline	34
Wheat	33
Peanut Butter	29
Micropore T...	27
Hay Fever	25
Strawberries	25
Almonds	23

Result 72

Output

Allergy Prevalence & Clinical Safety

- Supply Chain Risk: Identifies which alternative medications (non-allergic substitutes) must be kept in high inventory to avoid treatment delays.
- Environmental Exposure Risk: High-frequency food or material allergies (Latex/Nuts) trigger mandatory hospital-wide "Safe Zone" protocols.
- Emergency Readiness: Maps the most likely "Anaphylactic Triggers," ensuring the ER is stocked with the specific life-saving drugs needed for the local population.

Queries: 26

```
54 •  select first_name, last_name, birth_date from patients
55    where birth_date between '1970-01-01' and '1979-01-01' order by birth_date asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	birth_date
Frances	Kobayakawa	1970-01-02
Sunny	Burrell	1970-01-07
Penelope	Beckett	1970-01-14
Deborah	Stewart	1970-01-14
Augusta	Decker	1970-01-22
Sookie	Brearly	1970-02-01
Temple	Wylie	1970-02-10
Deanna	Spano	1970-03-23
Jadu	Principal	1970-03-28
Betty	Stephens	1970-03-28
Jo	Sahid	1970-03-31
Santino	Hawker	1970-04-08
Andy	Guarnaccia	1970-04-10
Richard	Arroway	1970-04-13
Alan	Davis	1970-04-14
Cameron	Spellman	1970-04-21
Bertha	Smart	1970-05-06

patients 73

Output

The use of this query

- Decade Filtering
- Insight: The line chart shows you if your patient population is mostly elderly or younger, which is a key metric in medical reporting.
- Chronological Order

Queries: 27

```
select concat(upper(last_name),' , ', lower(first_name)) as formatted_full_name
from patients order by first_name desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

formatted_full_name
MILLER , zoe
CORBIE , ziva
KOBAYAKAWA , zenigata
OVERSTREET , zenigata
BENNETT , zen
MEPHESTO , zelda
MORRIS , zelda
THOMAS , zefram
FLUTE , zefram
MARONEY , zatanna
TYRELL , zatanna
CHE , zane
WONG , zane
DREW , zack
BAKSHI , zachary
PRESTON , yusuke
LAZARUS , yuri

Result 74

Output

Report for query 27

- I used

Queries: 28

```
61 •  select province_id, sum(height) as total_height
62    from patients group by province_id having total_height>=7000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

province_id	total_height
BC	7720
NS	9765
ON	678037

Result 78

Output

Cont..

this type of query is often used for **Resource Allocation** or **Regional Analytics**:

- Physical Capacity
- Data Auditing

Queries: 29

```
63 •  select (max(weight)-min(weight)) as weight_diffrence from patients where last_name like 'Maroni%';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	weight_diffrence
▶	71

Result 82 ×

Output

The benefit of this query

- Identifying Data Entry Anomalies: (e.g., entering 700 instead of 70.0).
- Pediatric vs. Adult Care Tracking: They ensure pediatric scales (for infants) and standard scales (for adults) are both calibrated in the same wing.
- Monitoring Familial Health Trends (Genetics)
- Patient Progress (The "Weight Loss/Gain" Audit)

Queries: 30

```sql
select day(admission_date) as day_of_month, count(*) as admission_total from admissions group by day_of_month
order by admission_total desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

day_of_month	admission_total
4	184
11	184
9	183
2	180
12	179
6	179
16	177
21	174
13	173
28	173
7	167
8	166
18	166
3	166
20	165
30	165
10	164

Result 87

Cont..

Identifying these "Peak Days" is crucial for Operational Risk Management:

- **Staffing Exhaustion Risk**
- **The "Payday Effect" (UAE Context): This creates a Cyclical Capacity Risk.**
- **Emergency Room (ER) Bottleneck**

Queries: 31

```
select day(admission_date) as day_of_month, count(*) as admission_total from admissions group by day_of_month
order by admission_total desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

day_of_month	admission_total
4	184
11	184
9	183
2	180
12	179
6	179
16	177
21	174
13	173
28	173
7	167
8	166
18	166
3	166
20	165
30	165
10	164

Result 87

Weight Distribution & Clinical Risk Analysis

- The Metric: Patients segmented into 10kg cohorts to identify population-level obesity trends.
- Dosing & Pharmacy Risk: Highlights the need for weight-adjusted pharmacy protocols to prevent ineffective treatment in higher weight groups.
- Infrastructure Planning: Provides data-driven evidence for purchasing bariatric-grade furniture and diagnostic imaging tables (MRI/CT) that can support higher weight limits.

Queries: 33

```
68 •  select patient_id, weight, height,
69    case
70        when (weight/(height/10)*(height/10))>=30 then 1 else 0 end as isObese from patients;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

patient_id	weight	height	isObese
1	65	156	1
2	76	185	1
3	106	194	1
4	104	191	1
5	10	47	0
6	5	43	0
7	117	180	1
8	105	174	1
9	95	173	1
10	61	157	1
11	74	158	1
12	46	145	1
13	77	146	1
14	95	220	1
15	72	172	1
16	59	153	1
17	114	179	1

Result 90

Obesity Prevalence & Chronic Disease Risk

- **The Metric: Boolean identification of patients with $BMI \ge 30$.**
- **Predictive Value: Acts as a "Primary Indicator" for the development of chronic conditions like Type 2 Diabetes and Hypertension.**
- **Preventative Strategy: Allows the hospital to target "isObese" patients for early intervention screenings, reducing long-term hospital admission costs.**

Queries: 33

```
71 •  select p.patient_id, p.first_name, p.last_name,
72    d.specialty from patients p join admissions a on p.patient_id=a.patient_id
73    join doctors d on a.attending_doctor_id = d.doctor_id WHERE a.diagnosis = 'Epilepsy' AND d.first_name = 'Lisa';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_id	first_name	last_name	specialty
468	Frank	Anderson	Obstetrician/Gynecologist
701	Precious	Ashton	Obstetrician/Gynecologist

Result 92

Output

Data analyst competencies

- Clinical Quality Assurance: You are verifying that a "Neurology-level" condition (Epilepsy) is aligned with the appropriate specialist
- Operational Risk Management: By identifying specific doctor-patient assignments, you help prevent "Adverse Drug Events" or treatment omissions that can occur if a patient is seen by the wrong department.
- Data Integrity

Strategic Recommendations for Clinical & Operational Optimization

- Implement Automated Data Validation: To mitigate "Physical Risk" (e.g., the 700kg entry error), deploy SQL triggers or Python validation scripts that flag outliers in weight and height during the data entry phase.
- Dynamic Resource Allocation: Use "Peak Day" and "Payday Effect" insights to adjust staffing levels in the ER and outpatient clinics, ensuring high-demand periods are sufficiently covered to prevent "Staffing Exhaustion.“
- Targeted Preventative Programs: Leverage the BMI $\ge$ 30 (isObese) identification to enroll high-risk patients in early-intervention wellness programs, reducing long-term "Readmission Risk" and associated hospital costs.
- Pharmacy Safety Protocols: Standardize "NKA" (No Known Allergy) entries across the database to ensure every patient record provides clear, actionable safety data for pharmacists, reducing "Adverse Drug Events."

Conclusion

Transforming Raw Data into Healthcare Intelligence

- **Integrated Data Ecosystem: This project demonstrates that by successfully joining Patient, Admission, and Doctor tables, we can move from isolated data points to a holistic "360-degree view" of patient care.**
- **Risk-Centric Analysis: The transition from simple queries to "Risk Factor" identification proves that SQL is a powerful tool for predicting clinical outcomes and financial sustainability in a healthcare setting.**

Cont..

- Operational Excellence: By identifying "High-Utilizers" and "Same-Day Admissions," we provide hospital administration with the evidence needed to improve patient flow and departmental efficiency.
- Final Statement: The Medical Data History project successfully bridges the gap between database management and actionable healthcare strategy, providing a scalable framework for improving both patient safety and hospital performance.

END.